# Medical term simplifer

Module info , models and pipelines

# Icebrkr
# Module Info

# Medical Term Simplifier

*Bridging Clinical Language and Patient Understanding*

## Problem

Medical reports are often filled with complex clinical terminology that patients struggle to understand. Additionally, written reports are inaccessible for users with low literacy or visual impairments.

## Our Module

- An AI-powered Medical Report Simplification module that:
- Accepts uploaded medical reports
- Extracts and understands clinical information
- Simplifies complex medical terminology
- Delivers explanations through text and accent-aware speech

## Example

- Input: Medical Report (PDF/Image/Text)
- Output: Simplified explanation + Voice output in the user's preferred language/accent

# Tech Stack

| Pipeline Stage | Technology / Stack | Purpose |
|---|---|---|
| **Document Processing** | PyMuPDF (fitz) for PDFs, Tesseract OCR or PaddleOCR (for scanned reports), OpenCV (image preprocessing) | Extract clean text from reports |
| **Text Preprocessing** | spaCy, NLTK, regex | Sentence segmentation, normalization, abbreviation handling |
| **Context Understanding** | **BioClinical ModernBERT** (Hugging Face Transformers) | Generate contextual embeddings and understand clinical semantics |
| **Medical Entity Detection** | **OpenMed PharmaDetect** | Detect diseases, drugs, procedures, symptoms, lab values, anatomy, etc. |
| **Prompt Construction** | Python + Jinja2/Prompt Templates | Combine extracted entities and report context into a structured prompt for the LLM |
| **Simplification Engine** | ibm-granite/granite-4.1-3b / Qwen/Qwen2.5-0.5B-Instruct | Generate patient-friendly explanations while preserving clinical meaning |
| **Translation** | AI4Bharat IndicTrans2 | Translate explanations into Indian languages |
| **Accent Selection** | AI4Bharat Project Vaani (accent metadata) | Determine regional pronunciation profile |
| **Speech Synthesis** | Indic Parler TTS / Coqui XTTS / IndicTTS | Generate natural speech in the selected language and accent |

# Production Pipeline

```
Medical Report
      │
      ├──────────────┐
      ▼              ▼
BioClinical        OpenMed
ModernBERT         PharmaDetect
(Context)          (Clinical Entities)
      └──────────────┘
             │
             ▼
Medical Prompt Construction
             │
             ▼
Granite / Qwen Generator
             │
             ▼
Simplified Medical Explanation
```

## ROLE OF EACH MODEL

- **BioClinical ModernBERT**
    a. Understands clinical context
    b. Generates contextual embeddings
    c. Handles long medical reports
- **OpenMed PharmaDetect**
    - Detects diseases, medications, procedures, lab values, and other medical entities
    - Ensures important clinical information is retained
- **Granite / Qwen**
    - Converts extracted medical information into natural, patient-friendly explanations
    - Produces coherent, conversational output

# Accent-Aware Communication

*After simplification, the response is converted into speech.*

```
Simplified Text
       |
       ▼
Language Translation


       |
       ▼
Accent Detection
       |
       ▼
Accent-aware TTS
       |
       ▼
Natural Speech Output
```

- **Role:** Personalized Voice Output
- **Possible Models**
  - Project Vaani – Regional Indian accents and dialects.
  - IndicTTS / XTTS / Coqui TTS – Multilingual speech synthesis.
- **Purpose**
  - Convert the simplified explanation into speech.
  - Produce pronunciation closer to the user's regional accent.
  - Improve accessibility and comprehension for multilingual users.

# Each step explained

**1. Document Processing**

**Input:**

PDF

Image

Text

**Tasks:**

OCR (only if scanned)

Remove headers/footers

Preserve report structure

Normalize whitespace

Sentence segmentation

**Output:**

Clean clinical text.

**2. BioClinical ModernBERT**

**Input:**

Clinical text

**Tasks:**

- Encode semantic meaning
- Understand sentence context
- Resolve abbreviations
- Understand negations

**Output:**

Dense contextual embeddings.

These embeddings capture meaning rather than individual words.

# EACH STEP EXPLAINED

**3. OpenMed PharmaDetect**

**Input:**

Clinical text

**Tasks:**

Entity Recognition

**Find**

- Diseases
- Symptoms
- Drugs
- Dosages
- Procedures
- Organs
- Lab Tests

Output

```
Disease:
  Diabetes Mellitus

Drug:
  Metformin

Procedure:
  MRI Brain

Lab Test:
  HbA1c
```

**4. Prompt Construction**

```
Context

Patient diagnosed with Type 2 Diabetes.

Detected entities

Disease:
Type 2 Diabetes

Medication:
Metformin

Lab Result:
HbA1c = 8.4

Task

Explain this report in simple English.
Maintain clinical accuracy.
Do not omit important findings.
```

Instead of sending the raw report to Granite, construct something like this

This dramatically improves LLM consistency.

# EACH STEP EXPLAINED

**5. Granite / Qwen**

**Consumes**

- embeddings
- extracted entities
- structured prompt

**Produces**

Natural explanation.

6. **Multilingual & Accent-Adaptive Speech Generation**

**Tech Stack**

- IndicTrans2 – Translate the explanation into the patient's preferred language.
- Project Vaani – Selects an appropriate regional accent or dialect profile.
- Indic Parler TTS / XTTS – Generates natural speech using the selected language and accent.

**Why this stage?**

- Makes reports accessible to multilingual users.
- Improves comprehension through familiar pronunciation.
- Supports regional dialects rather than only standard language.

# Final architecture

```
Report Upload
      │
      ▼
OCR & Document Processing
      │
      ▼
BioClinical ModernBERT ────┐
(Context Understanding)    │
                           ├──► Prompt Construction
OpenMed PharmaDetect ──────┘
(Entity Extraction)
      │
      ▼
Granite / Qwen
(Explanation Generation)
      │
      ▼
Clinical Consistency Validation
      │
      ▼
IndicTrans2 (Translation)
      │
      ▼
Project Vaani
(Accent Profile)
      │
      ▼
Indic Parler TTS / XTTS
      │
      ▼
Patient
```